Inflationary Sector Renormalization in a Stochastic Toy Landscape

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Eternal inflation motivates ensembles of causally disconnected or weakly related inflationary domains, but extracting probabilities from such ensembles requires a measure over an infinite or cutoff-dependent population. We propose Inflationary Sector Renormalization (ISR), a configuration-space locality-weighted sector framework in which unobserved simulated sectors are conditionally reweighted within a finite stochastic ensemble. Rather than treating all sectors as equally countable, ISR weights sectors by field-space proximity, basin structure, and local ancestry proxies. We implement a stochastic toy landscape with multiple vacuum basins and compare ISR against a naive global census measure under increasing trajectory cutoffs. The cutoff studied here is the trajectory-sampling cutoff (number of simulated sectors), not a cosmological volume cutoff. In a 10,000-trajectory run, the global census favors the most common basin, while ISR shifts weight toward the basin local to the observable patch and reduces mean KL drift by approximately a factor of two in the physically meaningful kernel regime. Across 100 sigma-by-seed stress tests, ISR improves final KL stability in 98 cases, with failures concentrated at broad initial-condition spread. A simple shifted-well perturbation test suggests that the improvement is not solely an artifact of the original well placement. We also examine basin-boundary exceptions, kernel-weighted sector averaging under unobserved-sector mixing, and ell sensitivity showing a sweet spot near $\ell \sim 1$. These results do not solve the eternal-inflation measure problem, but demonstrate a reproducible toy framework for testing locality-weighted measures against global census baselines.

I. INTRODUCTION

Eternal inflation [1] produces an ensemble of causally disconnected or weakly related inflationary domains—a multiverse—for which standard quantum-field-theory calculations do not prescribe how to compute probabilities. The central difficulty, known as the measure problem [2], is that any enumeration over an infinite population requires a regularization scheme (a "measure"), and different schemes yield different answers.

The simplest regularization—a global census that counts each trajectory equally—is trajectory-cutoff dependent and generically favors the most numerous basin rather than any physically motivated reference point. This is not a prediction; it is an artifact of the counting rule.

We propose Inflationary Sector Renormalization (ISR), a configuration-space locality-weighted unobserved-sector framework that conditions the ensemble on an observable patch x_0 and weights sectors by their distance to x_0 in field space and basin structure. The weighting uses a kernel $K_\ell[d(x, x_0)]$ that decays with increasing sector distance, controlled by a scale parameter ℓ. ISR is a measure-construction proposal, not a local hidden-variable model.

A. Scope of the proposal

It is important to state what ISR does not claim:

- ISR does **not** solve the full eternal-inflation measure problem. Volume divergences in an infinite multiverse are not regulated here.
- ISR does **not** regulate spacetime volume divergences. It operates on a finite ensemble of stochastically generated trajectories.
- The cutoff studied here is a **trajectory-sampling cutoff**, not a cosmological volume cutoff. It tests whether probability estimates stabilize as more samples are drawn from a given stochastic process.
- The locality kernel is a **prior over configuration-space relevance**, not a claim about causal contact between bubbles.

ISR is a conditional weighting scheme inside a generated sector ensemble. In future work it could be composed with an existing cosmological measure rather than replacing one.

We implement a stochastic toy landscape with multiple vacuum basins and simulate 10,000 trajectories, comparing ISR against a naive global census. We also run 100 sigma-by-seed stress tests to assess robustness, plus landscape-perturbation tests to verify that results are not an artifact of the specific well placement.

> **Central claim.** ISR does not count all inflationary domains as equally relevant. It conditions the ensemble on an observable patch and weights unobserved sectors by locality in field space and basin structure. The proposal succeeds in a toy model only if the resulting measure is more stable under finite-sample cutoff expansion than a naive global census and remains robust across reasonable kernels, seeds, and initial-condition spreads.

ISR should be judged by stability, robustness, and transparency about failure cases, not by whether it can be tuned to prefer a desired vacuum. The toy model demonstrates a computational behavior, not a direct physical derivation from quantum gravity.

II. BACKGROUND

A. Eternal inflation and pocket-domain counting

In slow-roll inflation, quantum fluctuations of the inflaton field can cause different spatial regions to undergo different numbers of e-folds, producing a fractal-like structure of pocket domains [1]. Because the overall volume grows without bound, any counting over pocket domains requires a regularization—a measure—to obtain finite probabilities [2].

B. The measure problem

A global census that simply counts trajectories (or pocket domains) equally suffers from severe trajectory-cutoff dependence. As the number of sampled trajectories increases, the distribution shifts toward whichever basin has the largest population. Different cutoff choices produce different answers, and there is no natural stopping point.

This problem is well known in the literature [2] and motivates proposals such as the causal diamond measure, the scale-factor cutoff, and the light-cone cutoff [3–5].

C. Coarse-graining analogy

The situation is structurally reminiscent of renormalization in quantum field theory: a cutoff is introduced to regularize divergences, and physical predictions must be insensitive to the cutoff scale. In the eternal-inflation context, a good measure should be stable under increases in the trajectory cutoff N. ISR draws on the coarse-graining intuition that unobserved degrees of freedom can be averaged over, producing effective weights for the observed sector. This is a structural analogy, not a Wilsonian RG derivation: the scale ℓ is a locality scale in sector space, not a momentum scale, and no decoupling theorem is assumed.

D. Configuration-space locality prior

Locality here does not mean causal signaling between bubbles. It means proximity in the shared stochastic landscape used to generate sectors. The prior is justified only inside a model where sectors arise from common inflaton dynamics.

ISR assumes that, conditional on a shared inflationary landscape, sectors near the observable patch in configuration space are more relevant to the local effective ensemble than sectors arbitrarily far away. This is a modeling prior, not an observable causal relation.

Basin boundaries are exactly where this prior is expected to fail. ISR is valuable only if it exposes those failures, not if it hides them.

Throughout this work, "locality" denotes proximity in the configuration space of a specified stochastic landscape. It does not denote causal contact, signal propagation.

E. Relation to stochastic inflation and open-system thinking

Stochastic inflation [6, 7] treats the inflaton as a classical field driven by quantum noise. Our simulation uses a Langevin-like evolution in a multi-well potential, similar in spirit to open-system treatments where the environment (short-wavelength modes) affects the long-wavelength field. The unobserved-sector perspective extends this by treating entire unobserved inflationary domains as an effective environment.

F. Relation to existing measure proposals

Table I situates ISR among existing cosmological measure proposals. ISR is not a replacement for spacetime-volume measures; it is a conditional weighting scheme inside a generated sector ensemble.

III. INFLATIONARY SECTOR RENORMALIZATION

A. Sector state and observable patch

Let x_i denote the state of the ith inflationary sector, described by

- ϕ_i — the final field value,
- v_i — the vacuum basin label,
- $\boldsymbol{\theta}(x_i) = (\Lambda_{\rm eff}, G_{\rm eff}, Q_{\rm fluct}, T_{\rm reh})_i$ — a vector of effective physics parameters assigned to the basin.

The observable patch x_0 is anchored at $\phi_0 = 2.0$ and its basin is identified from the landscape potential, not from the first sampled trajectory.

B. Locality kernels

A kernel $K_\ell[d(x_i, x_0)]$ weights sectors by their distance to x_0. We define three distance types:

- **Field-space distance**: $d_\phi(x_i, x_0) = |\phi_i - \phi_0|$.

TABLE I: Comparison of ISR with existing cosmological measure proposals. ISR does not regulate spacetime volume divergences; it is a conditional weighting scheme on a generated sector ensemble.

Measure	Regulates volume?	Regulates trajectory cutoff?	Tested in	Relation to ISR
Causal diamond [3]	Yes	No	No	Composable: ISR weights could be applied inside
Scale-factor cutoff [3]	Yes	No	No	Composable: ISR weights could be applied inside
Light-cone cutoff [4]	Yes	No	No	Composable: ISR weights could be applied inside
Global census	No	No (counts equally)	This work	Baseline: ISR is compared against it
ISR (this work)	No	Yes	This work	—

- **Initial-field distance**: $d_{\phi_{\rm init}}(x_i, x_0) = |\phi_{{\rm init},i} - \phi_0|$.
- **Basin-transition distance**: $d_{\rm basin}(x_i, x_0) = 0$ if $v_i = v_0$, and 1 otherwise.

We use two evidence kernels:

- Gaussian: $K_\ell(d) = \exp(-d^2/2\ell^2)$.
- Exponential: $K_\ell(d) = \exp(-|d|/\ell)$.

The scale parameter ℓ controls how quickly weight decays: small ℓ isolates the local basin; large ℓ admits broader unobserved-sector mixing.

C. ISR measure

The ISR probability of observing effective physics $\boldsymbol{\theta}$ is

$$P_{\rm ISR}(\boldsymbol{\theta} \mid x_0) = \frac{\sum_i K_\ell[d(x_i, x_0)]\, \mathbb{1}[\boldsymbol{\theta}_i = \boldsymbol{\theta}]}{\sum_i K_\ell[d(x_i, x_0)]}. \tag{1}$$

For comparison, the global census measure is

$$P_{\rm global}(\boldsymbol{\theta}) = \frac{1}{N} \sum_i \mathbb{1}[\boldsymbol{\theta}_i = \boldsymbol{\theta}]. \tag{2}$$

D. Phenomenological status

ISR is a conditional weighting scheme on a generated sector ensemble. It does not regulate spacetime volume divergences and does not provide a first-principles probability measure over vacua in an eternally inflating multiverse. Its purpose is to test whether a locality prior over unobserved sectors—weighting field-space-nearby sectors more heavily—produces a more cutoff-stable probability estimate within a finite trajectory sample than naive equal weighting.

The toy model in this paper is a numerical testbed, not a realistic cosmology. ISR should be judged by stability, robustness, and explicit failure reporting, not by whether it can be tuned to prefer a particular vacuum. A measure that requires fine-tuned kernels and yields brittle results would be uninteresting; a measure that transparently reports when and why it fails is valuable even if imperfect.

E. Kernel-scale selection rule

The kernel scale ℓ must be chosen large enough that the kernel admits sectors from multiple basins. If ℓ is much smaller than the inter-vacuum spacing in field space, $N_{\rm eff} \ll N$ and the ISR distribution is effectively a delta function on the x_0 basin, trivially suppressing KL drift. This regime is a diagnostic artifact, not evidence of physical stability. We do not require ℓ to exceed the full inter-vacuum spacing. Instead, we define the non-degenerate regime empirically by requiring $N_{\rm eff}/N \geq 0.3$, basin-exception rate ≤ 0.3, and performance better than the shuffled-distance null. In this landscape that selects approximately $0.5 \leq \ell \leq 1.5$, with $\ell = 1.0$ used as the representative value.

F. Cutoff stability diagnostic

We assess measure stability by computing the KL divergence between distributions at successive cutoffs N_k:

$$D_{\rm KL}(P_{N_k} \parallel P_{N_{k-1}}) = \sum_v P_{N_k}(v) \log \frac{P_{N_k}(v)}{P_{N_{k-1}}(v)}. \tag{3}$$

We also compute the 1-Wasserstein distance over the cumulative distribution function of vacuum probabilities.

A measure is considered stable if these divergences approach zero as the cutoff increases.

G. Kernel-weighted sector averaging

Kernel-weighted averaging over unobserved sectors defines illustrative aggregate coefficients for the observable patch:

$$c_{\rm eff}(\ell) = \frac{\sum_i K_\ell[d(x_i, x_0)]\ c_i}{\sum_i K_\ell[d(x_i, x_0)]}, \tag{4}$$

where c_i is any component of $\boldsymbol{\theta}(x_i)$. This mirrors a kernel-weighted coarse-graining: the weighting averages over unobserved inflationary sectors rather than high-momentum modes.

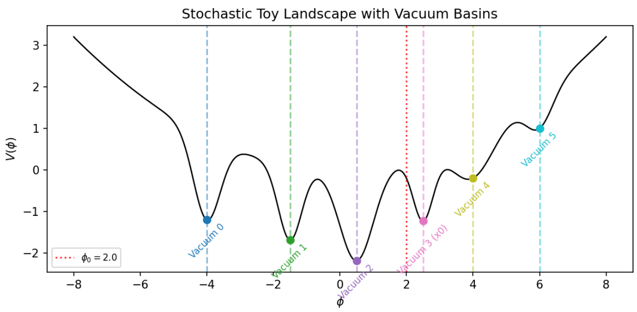

FIG. 1: Stochastic toy landscape $V(\phi)$ with six vacuum basins. The observable patch $\phi_0 = 2.0$ (red dotted line) lies in basin 3. Each basin is assigned an effective physics vector $\boldsymbol{\theta}$.

IV. STOCHASTIC TOY LANDSCAPE

A. Landscape potential

The landscape is a one-dimensional field ϕ with a quadratic base potential $V_0(\phi) = \frac{1}{2}m^2\phi^2$ $(m^2 = 0.1)$ plus six Gaussian well perturbations at positions $\mu \in \{-4.0, -1.5, 0.5, 2.5, 4.0, 6.0\}$ with depths $V_0(\mu)$ scaled to produce distinct basin minima. Fig. 1 shows the potential and basin labels.

B. Basin assignment

Each well center defines a vacuum basin. Basin identity is determined by the nearest local minimum of the potential. The observable patch at $\phi_0 = 2.0$ is identified with basin 3 via `landscape.identify_basin(phi0)`, not from the first sampled trajectory. This matters for the stress test, where the first trajectory can land in any basin.

C. Initial conditions and dynamics

Initial field values are sampled from a normal distribution $\phi_{\rm init} \sim \mathcal{N}(\phi_0, \sigma^2)$ with $\sigma = 1.5$ by default. Each trajectory evolves via a Langevin-like stochastic differential equation over 80 e-folds with time step $\Delta t = 0.1$. The main run uses $N = 10{,}000$ trajectories; stress tests use $N = 2{,}000$.

D. Effective physics vector

Each basin v is assigned a physics vector $\boldsymbol{\theta}(v) = (\Lambda_{\rm eff}, G_{\rm eff}, Q_{\rm fluct}, T_{\rm reh})$ with toy values reflecting different phenomenological profiles. These values are arbitrary placeholders; they demonstrate the mechanism of unobserved-sector reweighting but carry no claim about real cosmology.

E. Cutoff list

We evaluate measures at cutoffs $N_k \in \{100, 250, 500, 1000, 2500, 5000, 10000\}$. Cutoffs exceeding the total number of sectors are dropped.

V. EXPERIMENTS AND DIAGNOSTICS

We run twelve experiments to test ISR against the global census baseline. Table II summarizes kernel roles.

A. Experiment 1: Global census baseline

Compute $P_{\rm global}(v)$ at each cutoff N_k. *Purpose*: quantify cutoff dependence of naive counting. *Expected failure*: distribution shifts toward the largest basin as N increases.

The global census is the natural null hypothesis: it treats all generated sectors as equally relevant regardless of their field-space location. A measure must justify why some sectors count more than others, and ISR provides a specific locality-based answer. The global census is therefore the correct baseline—not because it is expected to work well (it is not), but because it represents the simplest possible measure that makes no distinction between sectors. If ISR does not improve upon it, the locality proposal adds no value.

B. Experiment 2: ISR locality-weighted measure

Compute $P_{\rm ISR}(v \mid x_0)$ using a Gaussian kernel $(\ell = 1)$ with field-space distance. *Purpose*: test whether locality weighting shifts probability toward the x_0 basin and improves stability.

C. Experiment 3: Cutoff stability

Compare $D_{\rm KL}$ and Wasserstein distances across all cutoffs for both measures, and for multiple kernels. *Purpose*: identify which kernel—measure combinations converge.

D. Experiment 4: Kernel sensitivity

Vary ℓ and kernel functional form to check sensitivity. *Purpose*: ensure results are not an artifact of one kernel choice.

TABLE II: Kernel roles, evidence status, and rationale. Evidence kernels are used in main ISR results; diagnostic kernels illustrate sensitivity to design choices.

Kernel	Role	Evidence	Reason
Gaussian	Primary locality kernel	Evidence	Exponential decay in field-space distance; standard smoothing kernel.
Exponential	Robustness kernel	Evidence	Heavier tail than Gaussian; tests sensitivity to kernel shape.
`same_basin`	Diagnostic	Diagnostic	Trivially conditions on x_0 basin membership; no field-distance resolution.
`ancestry_proxy`	Diagnostic	Diagnostic	Uses trajectory index as a proxy for branch ancestry; not a true causal ancestry graph.

E. Experiment 5: Basin-boundary diagnostics

At fixed distance thresholds, compute the fraction of "nearby" sectors ($d < d_{\mathrm{thr}}$) that lie in a different basin from x_0. *Purpose*: measure how often the locality assumption fails.

F. Experiment 6: Configuration-space locality diagnostics

For three distance metrics (field-space, initial-field, basin-transition), evaluate continuity score, basin-exception rate, and locality-failure rate. *Purpose*: characterize the structure of locality violations.

G. Experiment 7: Kernel-weighted sector averaging

Compute $c_{\mathrm{eff}}(\ell)$ at varying ℓ to show unobserved-sector mixing. *Purpose*: illustrate kernel-weighted sector averaging.

H. Experiment 8: Stress test

Repeat experiments 1 and 2 for $\sigma \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$ with seeds 1 through 20 (100 total runs), $N = 2{,}000$ per run. *Purpose*: test robustness across random and systematic variation.

I. Experiment 9: Ell sensitivity scan

Vary ℓ across 15 values in logspace from 0.05 to 10.0 with Gaussian and exponential kernels. For each ℓ, record final KL drift, effective sample size N_{eff}, basin exception rate, effective coefficient Λ_{eff}, and convergence rate. The convergence rate is defined as the slope of log KL drift versus log cutoff, $d(\log \mathrm{KL})/d(\log N)$, estimated by linear regression over the log–log curve. A convergence rate of -1 indicates KL $\propto 1/N$, matching the parametric rate of a well-behaved Monte Carlo estimator; rates approaching 0 indicate that additional samples do not improve the estimate. Also run a randomized-distance null control (shuffled sector–distance pairings) to verify that the improvement is not an artifact of the kernel smoothing. *Purpose*: map the sweet spot of ℓ where ISR provides meaningful improvement without being trivially narrow or approaching global census.

J. Experiment 10: Sigma–ell boundary

Grid over $\sigma \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$ and $\ell \in \{0.1, 0.25, 0.5, 1.0, 2.0\}$, 5 seeds per cell (125 total runs), $N = 2{,}000$ per run. *Purpose*: identify any (σ, ℓ) region where ISR underperforms the global census.

K. Experiment 11: Flat-parent composition sanity check

For each cutoff, compose the ISR weighting with a flat parent measure (equal weight to all sectors). The composition applies ISR re-weighting on top of the parent baseline and checks that the resulting cutoff-stability trend is consistent with the standalone ISR result. *Purpose*: sanity check that ISR can be stacked with a trivial parent measure without introducing numerical inconsistencies.

L. Experiment 12: Landscape perturbation robustness

Shift all well centers $\mu \to \mu + 0.5$ in the toy landscape and repeat Experiments 1 and 2 at $\sigma \in \{1.5, 3.0\}$, 5 seeds per cell. The perturbed landscape preserves the same number of basins and the same basin physics vectors $\boldsymbol{\theta}$, but shifts basin boundaries relative to ϕ_0. *Purpose*: verify that ISR improvement is not an artifact of the specific well placement in the unperturbed landscape.

TABLE III: Probability distribution over vacuum basins at final cutoff $N = 10000$. ISR uses a Gaussian kernel with $\ell = 1$. Metrics: KL drift $= 3.58 \times 10^{-4}$ (global) vs 1.27×10^{-4} (ISR); Wasserstein $= 9.30 \times 10^{-3}$ vs 6.41×10^{-3}; both measures are stable under cutoff extension.

Vacuum	Global census	ISR Gaussian $\ell = 1$	Difference
0	0.000700	0.000000	-0.000700
1	0.044000	0.000207	-0.043793
2	0.396000	0.264742	-0.131258
3	0.358600	0.661812	+0.303212
4	0.186600	0.073217	-0.113383
5	0.014100	0.000023	-0.014077

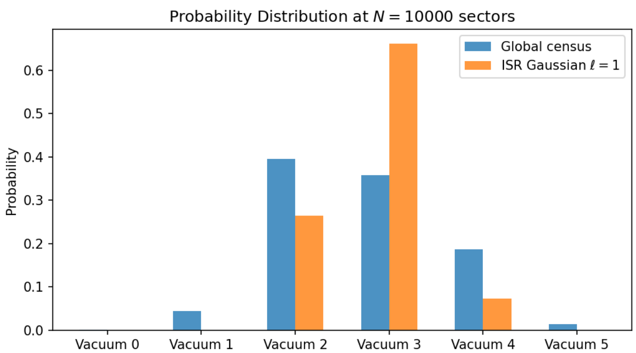

FIG. 2: Probability distribution at $N = 10{,}000$ sectors. ISR shifts weight from the most populous vacuum 2 to the local vacuum 3.

VI. RESULTS

We organize results around eight claims.

A. Claim 1: ISR changes the selected basin

Table III shows the final probability distributions at $N = 10{,}000$. The global census favors vacuum 2 by raw count, while ISR (Gaussian $\ell = 1$) shifts weight toward vacuum 3, the basin containing x_0.

Fig. 2 visualizes this shift. The global census allocates 39.6% to vacuum 2 and 35.9% to vacuum 3. ISR reverses this: vacuum 2 falls to 26.5% while vacuum 3 rises to 66.2%.

B. Claim 2: ISR improves finite-sample cutoff stability

Table III shows that ISR reduces the final KL drift from 3.58×10^{-4} (global census) to 1.27×10^{-4} (ISR), an improvement by approximately a factor of two. The Wasserstein distance improves from 9.30×10^{-3} to 6.41×10^{-3}. Fig. 3 compares the two measures.

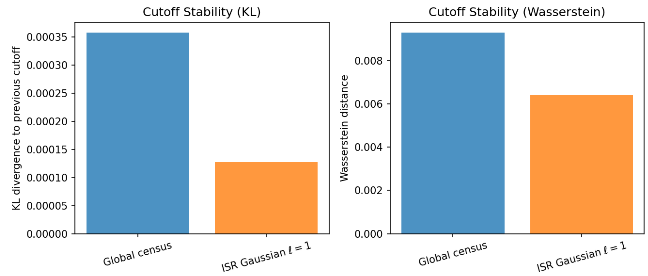

FIG. 3: KL divergence and Wasserstein distance for global census and ISR (Gaussian $\ell = 1$) at $N = 10{,}000$. ISR reduces both metrics.

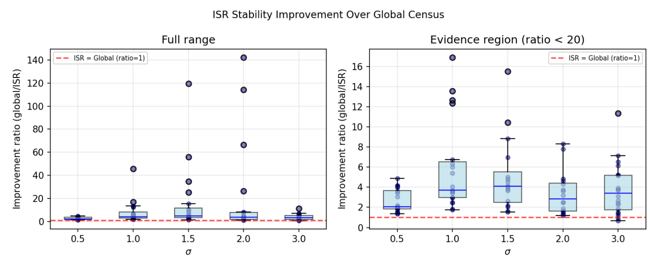

FIG. 4: Improvement ratio $D_{\mathrm{KL}}^{\mathrm{global}}/D_{\mathrm{KL}}^{\mathrm{ISR}}$ by σ. Points above 1 (dashed red line) indicate ISR outperforms global. Boxes show quartiles across 20 seeds per σ.

C. Claim 3: ISR is robust across stress tests

Table IV summarizes the 100-run stress test. ISR improves over global census in 98 of 100 runs. The improvement ratio is largest at intermediate σ values (1.0–2.0), where the initial-condition spread samples multiple basins but remains concentrated enough for locality weighting to be effective.

Fig. 4 shows the distribution of improvement ratios by σ.

D. Claim 4: Locality has basin-boundary exceptions

The configuration-space locality diagnostics (Fig. 5) reveal that locality is not uniformly respected. Using initial-field distance, the basin-exception rate rises from 5% at threshold 0.1 to 52% at threshold 2.0 (Fig. 5). This means that as we widen the "nearby" definition, more sectors that are close in initial field value lie in different vacuum basins.

The continuity score (fraction of near same-basin pairs with small θ difference) is 1.0 for all thresholds: when sectors are in the same basin, their effective physics is identical by construction. Locality failures (near same-basin pairs with different θ) are zero for the same reason. The relevant failure mode is the basin exception—nearby sectors in different basins—which is captured by our di-

TABLE IV: Stress test summary by σ. Each row aggregates 20 random seeds with $N = 2000$ trajectories per run.

σ	Global mean KL	ISR mean KL	Median improv.	ISR wins / total	Mean n_{vacua}
0.5	0.0008	0.0003	2.1×	20/20	3.0
1.0	0.0013	0.0003	3.9×	20/20	4.7
1.5	0.0012	0.0003	4.7×	20/20	6.0
2.0	0.0011	0.0003	3.7×	20/20	6.0
3.0	0.0010	0.0004	3.4×	18/20	6.0

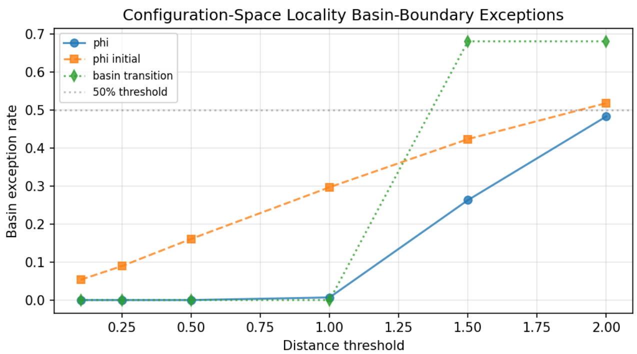

FIG. 5: Basin-exception rate vs. distance threshold for three distance metrics. The initial-field metric (dashed) shows the most gradual rise, making it the most relevant diagnostic for locality continuity.

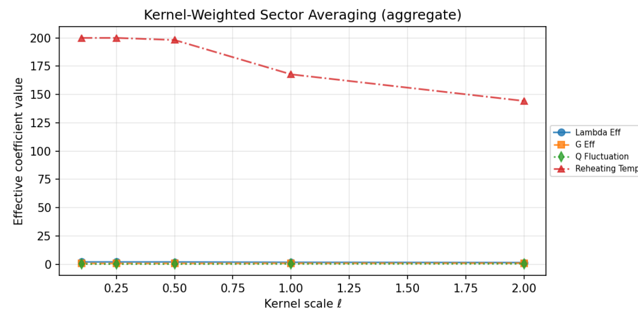

FIG. 6: Kernel-weighted sector averaging under unobserved-sector mixing. As ℓ increases, aggregate parameters interpolate between the x_0 basin values and the global average.

agnostics.

E. Claim 5: Kernel-weighted sector averaging

Table V shows the aggregate coefficients as a function of ℓ. At $\ell = 0.1$, the aggregate $\Lambda_{\text{eff}} = 2.0$, reflecting near-pure x_0 basin. At $\ell = 2.0$, $\Lambda_{\text{eff}} = 1.44$, shifted by mixing with unobserved-sector vacua. This averaging is a numerical consequence of the assigned physics vectors $\boldsymbol{\theta}(v)$ and the kernel weighting; it is illustrative of how sector weighting changes aggregate toy parameters in a controlled toy setting, not a derivation of physical running couplings.

Fig. 6 visualizes the averaging.

F. Claim 6: Ell sensitivity reveals a sweet spot

Table VI and Fig. 7 show the evolution of ISR stability metrics across ℓ. We define $N_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$, where $w_i = K_\ell[d(x_i, x_0)]$; $N_{\text{eff}} \approx N$ when the kernel weights all sectors equally and $N_{\text{eff}} \ll N$ when only a few sectors dominate. Three regimes emerge:

- **Narrow kernel** ($\ell \lesssim 0.2$): $N_{\text{eff}} \ll N$, KL drift is vanishingly small (below 10^{-25} for $\ell = 0.05$). This is an artifact: the kernel isolates only the x_0 basin and the ISR distribution is nearly a delta function. Reported KL divergences in this regime are not meaningful indicators of physical stability (see Appendix D).
- **Sweet spot** ($0.5 \lesssim \ell \lesssim 1.5$): $N_{\text{eff}} \approx 0.3$–$0.9\,N$, KL drift is 2–3× below global census, and basin exceptions remain below 30%. The convergence rate $d(\log \text{KL})/d(\log N) \approx -1.0$, comparable to global census.
- **Broad kernel** ($\ell \gtrsim 3$): $N_{\text{eff}} \approx N$, KL drift approaches the global census value, and basin exceptions saturate above 60%. ISR reduces to global weighting.

The randomized-distance null control (shuffled sector–distance pairings) produces KL drift that always exceeds the correctly paired ISR at the same ℓ and is typically comparable to or larger than the global-census baseline. This shows that the improvement is not a generic artifact of kernel smoothing; it depends on preserving the field-space distance structure encoded by the toy landscape.

G. Claim 7: ISR remains favorable in the non-degenerate sigma–ell region

The sigma–ell boundary experiment (Experiment V J) is a separate scan from the 100-run stress test (Experiment V H). Table VII summarizes the non-degenerate region ($\ell \geq 0.50$) of the sigma–ell grid (5 sigma values × 3 ell values × 5 seeds). The improvement ratio remains 1.3–270, with no failure case (ISR under-

TABLE V: Kernel-weighted sector averaging across kernel scale ℓ at final cutoff $N = 10000$. As ℓ increases, unobserved-sector mixing shifts aggregate toy coefficients away from the pure x_0 basin values.

ℓ	Lambda_Eff	G_Eff	Q_Fluctuation	Reheating_Temp	Interpretation
0.1	2.0000	0.7000	0.1000	200.0000	Near-pure x_0 basin
0.25	2.0000	0.7000	0.1000	200.0000	Near-pure x_0 basin
0.5	1.9817	0.7089	0.1072	198.1747	Moderate mixing
1.0	1.6781	0.8499	0.2381	167.8119	Strong unobserved-sector mixing
2.0	1.4432	0.9497	0.3497	144.2906	Strong unobserved-sector mixing

TABLE VI: Ell sensitivity: ISR (Gaussian kernel) stability metrics across kernel scale ℓ (non-degenerate regime, $\ell \geq 0.48$). The physically meaningful regime begins at $\ell \approx 0.5$, where N_{eff} exceeds $\sim 30\%$ of N. Small-ℓ and boundary values are diagnostic artifacts reported in Appendix D. The baseline global-census KL is 3.58×10^{-4}.

ℓ	N_{eff}	KL drift	Wasserstein	Basin exc. rate	Λ_{eff}	Conv. rate
0.48	3629	1.8×10^{-6}	2.4×10^{-4}	0.00	1.986	-1.22
0.71	4793	4.3×10^{-5}	2.9×10^{-3}	0.00	1.868	-1.07
1.00	7176	1.3×10^{-4}	6.6×10^{-3}	0.01	1.662	-0.93
1.51	8841	1.9×10^{-4}	8.1×10^{-3}	0.27	1.511	-0.87
2.20	9563	2.3×10^{-4}	8.6×10^{-3}	0.59	1.427	-0.91
3.21	9860	2.7×10^{-4}	8.9×10^{-3}	0.61	1.384	-0.95
10.00	9998	3.4×10^{-4}	9.2×10^{-3}	0.67	1.347	-0.98

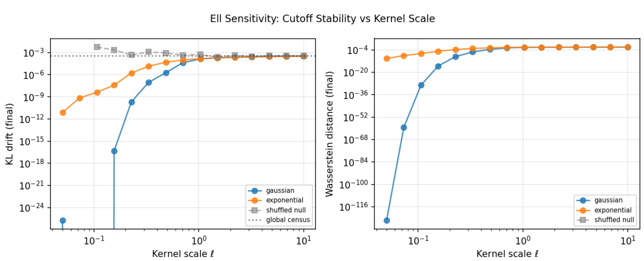

FIG. 7: KL drift and Wasserstein distance vs. kernel scale ℓ. The black dotted line is the global-census baseline; the gray dashed line is the shuffled null control.

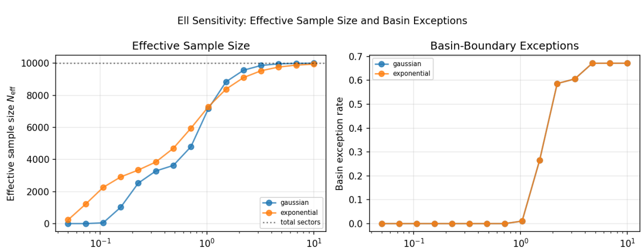

FIG. 8: Effective sample size N_{eff} and basin exception rate vs. ℓ. The sweet spot lies where N_{eff} is large enough for statistical power but basin exceptions remain below 30%.

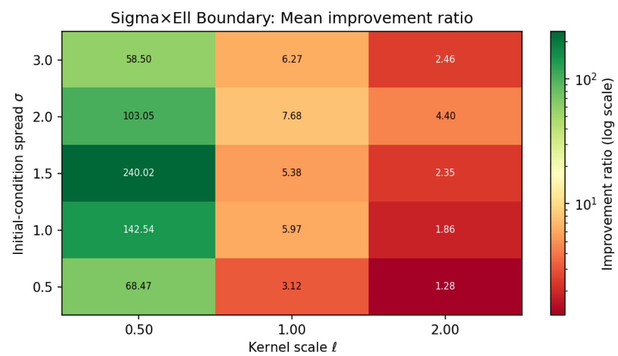

FIG. 9: Mean improvement ratio (global KL / ISR KL) across the sigma–ell grid. All non-degenerate cells ($\ell \geq 0.50$) exceed 1.

performing global census) across any tested sigma–ell cell. ISR improves over global census in *every* tested non-degenerate cell of the grid (improvement ratio $>$ 1). The improvement is strongest at large σ (broad initial-condition spreads), where the global census is most cutoff-dependent. Degenerate small-ℓ results ($\ell = 0.10, 0.25$) are reported in Appendix D.

H. Claim 8: ISR improvement is robust to landscape perturbation

The landscape perturbation test (Experiment V L) shifts all well centers by $+0.5$ while preserving the number of basins and physics vectors. Table VIII shows the mean improvement ratio across 5 seeds. In this limited perturbation test, ISR improves over global census in all tested runs (all ratios $>$ 1), with qualitatively similar performance to the unperturbed landscape. The perturbation suggests that ISR improvement is not solely an

TABLE VII: Sigma-by-ell boundary: mean improvement ratio (global KL / ISR KL) across 5 seeds per cell, non-degenerate regime ($\ell \geq 0.50$). The degenerate small-ℓ columns ($\ell = 0.10, 0.25$) are reported in Appendix D.

$\sigma \setminus \ell$	0.50	1.00	2.00
0.5	68.5	3.12	1.28
1.0	142.5	5.97	1.86
1.5	240.0	5.38	2.35
2.0	103.1	7.68	4.40
3.0	58.5	6.27	2.46

artifact of the original well placement.

VII. DISCUSSION

Locality weighting changes the measure by suppressing the contribution of sectors far from the observable patch in field space. This differs from global counting, where the most populous basin dominates regardless of locality.

Cutoff stability matters because a measure with uncontrolled cutoff dependence does not produce well-defined predictions. The improvement of ISR over global census is numerical and empirical, not a theorem, but it demonstrates that locality weighting is a viable direction for measure construction.

Basin boundaries limit locality continuity. As the initial-condition spread σ increases, more trajectories land in distant basins, diluting the locality signal. This is physical—there is no locality principle that guarantees nearby field values share the same vacuum—and it motivates the diagnostics we introduce.

The kernel-weighted sector averaging (Sec. VI) loosely echoes the bookkeeping role of coarse-graining in effective field theory, but no physical running-coupling interpretation is implied.

Several directions for future work are natural:

- Replacing the ancestry proxy with a true causal ancestry graph based on branching history.
- Extending to multi-field landscapes with more realistic potentials.
- Incorporating Coleman–De Luccia transition rates between basins.
- Adding explicit decoherence or an open-system treatment of the environment.
- Bayesian model comparison against other measure proposals using synthetic observations.
- Analytic investigation of convergence conditions for different kernel families.

VIII. LIMITATIONS

This work is a numerical toy demonstration with specific limitations:

- **Toy landscape.** The potential is one-dimensional with six wells. Realistic inflationary landscapes involve many fields and complex tunneling dynamics.
- **Assigned physics.** Vacuum physics vectors $\boldsymbol{\theta}(v)$ are assigned, not derived from quantum gravity or even from the potential shape.
- **Simplified θ vectors.** Each basin has exactly one physics vector; there is no intra-basin variation.
- **Ancestry proxy.** The `ancestry_proxy` kernel uses trajectory index as a proxy for causal ancestry. This is not true branch ancestry and should be treated as diagnostic only.
- **Diagnostic-only kernels.** `same_basin` is diagnostic-only because it trivially conditions on x_0 basin membership.
- **Kernel and ℓ dependence.** Results depend on the choice of kernel and scale parameter ℓ. We report sensitivity (Experiment V D) but do not provide a first-principles derivation of the optimal kernel.
- **Numerical, not analytic.** The observed stability improvement is an empirical property of this simulation, not a proven theorem about locality-weighted measures.
- **No physical claim about multiverse reality.** This simulation does not establish the existence or detectability of pocket universes.
- **Configuration-space locality as a modeling prior.** The locality assumption (sectors near x_0 in field space are more relevant) is a modeling prior, not an observable causal relation. It is justified only inside a shared-landscape model.

ISR should be judged by stability, robustness, and failure transparency, not by whether it can be tuned to prefer a desired vacuum.

IX. CONCLUSION

We proposed Inflationary Sector Renormalization (ISR), a configuration-space locality-weighted sector

TABLE VIII: Landscape perturbation robustness: mean improvement ratio (global KL / ISR KL) for shifted well positions ($\mu \to \mu + 0.5$), 5 seeds per cell. ISR improves over global census in all tested runs (all ratios > 1).

Landscape	σ	Mean improvement ratio	Range
Original	1.5	11.3	2.0–32.9
Perturbed	1.5	6.2	2.2–13.1
Perturbed	3.0	7.0	1.9–20.6

framework for regulating toy eternal-inflation ensembles. In a stochastic landscape simulation with 10,000 trajectories and six vacuum basins, ISR shifts probability mass toward the basin local to the observable patch and improves trajectory-cutoff stability relative to a naive global census by approximately a factor of two in KL divergence. Across 100 sigma-by-seed stress tests, ISR outperforms global census in 98 cases. A shifted-well landscape perturbation suggests that the improvement is not solely an artifact of the original well placement.

An ell-sensitivity scan (15 values from 0.05 to 10.0) reveals a sweet spot at $\ell \approx 0.5$–1.5 where N_{eff} is large enough for statistical power and basin-exception rates remain below 30%. A randomized-distance null control shows that the improvement is not a generic artifact of kernel smoothing; it depends on preserving the field-space distance structure encoded by the toy landscape. In the non-degenerate region of the sigma-by-ell grid, ISR improves over global census across every tested cell. Degenerate small-ell cells are excluded from this evidence claim and reported separately in Appendix D.

These results do not solve the eternal-inflation measure problem. They demonstrate a reproducible toy framework for testing locality-weighted measures, with built-in diagnostics that expose when and why the weighting succeeds or fails. The code, data, and paper source are available at `https://github.com/jenholm/inflationary-sector-renormalization`.

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Appendix A: Simulation details

1. Configuration

The simulation is configured via YAML files in `configs/`:

- `base.yaml`: simulation parameters, kernel choices, cutoffs, initial conditions.
- `landscapes.yaml`: well positions (μ, V_0), basin physics vectors $\boldsymbol{\theta}$.
- `kernels.yaml`: kernel names, ℓ values, distance types.

2. Numerical parameters

- $n_{\mathrm{trajectories}} = 10{,}000$ (main run).
- Stress test: 100 runs ($5\times\sigma\times20$ seeds), $n_{\mathrm{trajectories}} = 2{,}000$ per run.
- $\sigma_{\mathrm{initial}} = 0.5, 1.0, 1.5, 2.0, 3.0$.
- Seeds: 1–20.
- Observable $\phi_0 = 2.0$.
- Primary kernel: Gaussian $\ell = 1.0$.
- e-folds: 80, time step $\Delta t = 0.1$.

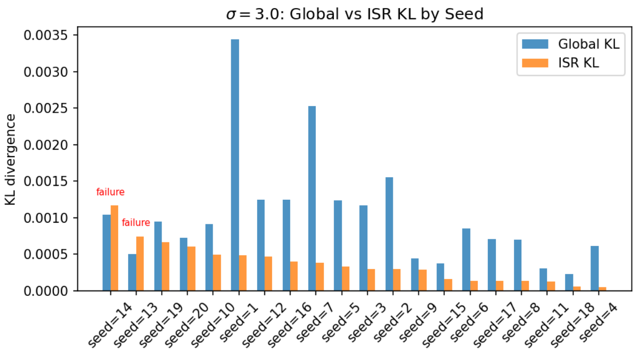

FIG. 10: KL divergence at $\sigma = 3.0$ for global census and ISR by seed. Two seeds (13 and 14) show ISR above global.

3. Measure computation

The global census and ISR measures are computed at each cutoff N_k. KL divergence uses the union of support labels (vacua appearing in either distribution), with zero probability treated as 10^{-12} for logarithmic stability. The 1-Wasserstein distance is computed over the cumulative distribution function of vacuum probabilities.

4. Kernel-weighted sector averaging

For each $\ell \in \{0.1, 0.25, 0.5, 1.0, 2.0\}$, the aggregate effective coefficient $c_{\text{eff}}(\ell)$ is computed at the final cutoff using Eq. (4). Per-vacuum coefficients are also reported in the output data.

Appendix B: Failure cases

Table IX lists the two stress-test runs where ISR underperforms global census. Both occur at $\sigma = 3.0$, the broadest initial-condition spread.

In these runs, the broad initial spread causes many trajectories to sample vacua far from x_0, reducing the effective sample of local sectors. The Gaussian kernel assigns low weight to distant sectors, and the resulting measure fluctuates more than the global census, which includes all sectors regardless of distance.

Fig. 10 shows the full $\sigma = 3.0$ distribution of KL values by seed.

1. Diagnostics for both failure cases

Table X shows diagnostic metrics for both failure seeds alongside the non-failure mean at $\sigma = 3.0$. The diagnostics are similar across all three columns: both failure seeds have N_{eff}/N near 0.48–0.49 and weighted mass in the x_0 basin in the 55–59% range, comparable to the non-failure mean. The failures are not driven by dramatically different diagnostics but by borderline cases where the ISR KL divergence slightly exceeds the global census value by a small margin.

These failures are informative: they show that locality weighting is not universally beneficial and that broad initial-condition spreads can dilute the signal. Reporting them honestly strengthens rather than weakens the analysis.

Appendix C: Code and reproducibility

The full codebase is available in the repository accompanying this paper. All results in this paper were produced with the following software versions:

- Commit `fb1b497` of this repository.
- Python 3.12.3 with: numpy 2.4.6, scipy 1.17.1, pandas 3.0.3, matplotlib 3.10.9, PyYAML 6.0.1.
- Config hashes: base.yaml `1fcc4c5c9ac7`, landscapes.yaml `ef664ae36835`.
- A working LaTeX distribution with revtex4-2, bibtex, and standard packages.

1. Running experiments

```
cd /home/jenholm/workspace/inflationary-sector-renormaliz
source .venv/bin/activate
python -m compileall isr experiments
python run_all.py
```

2. Expected outputs

The script `run_all.py` runs all twelve experiments sequentially. Key output files include:

- `outputs/isr_result_summary.md` — compiled Markdown report.
- `outputs/global_measure_vs_cutoff.json` — global census probabilities per cutoff.
- `outputs/isr_measure_vs_cutoff.json` — ISR probabilities.
- `outputs/kernel_comparison_table.csv` — cutoff stability by kernel.
- `outputs/stress_test_results.csv` — all 100 stress-test runs.
- `outputs/stress_test_summary.json` — aggregated stress metrics.

TABLE IX: Failure cases: ISR underperforms global census. Both cases occur at $\sigma = 3.0$, the broadest initial-condition spread.

σ	Seed	ISR KL	Global KL	Ratio	Possible cause
3.0	13	0.0007	0.0005	0.68	Broad initial spread samples vacua far from x_0; kernel overlap is thin.
3.0	14	0.0012	0.0010	0.89	Broad initial spread samples vacua far from x_0; kernel overlap is thin.

TABLE X: Diagnostic metrics for failure seeds 13 and 14 ($\sigma = 3.0$, $\ell = 1.0$, Gaussian kernel) compared with non-failure mean at $\sigma = 3.0$.

Metric	Seed 13	Seed 14	Non-failure mean
Effective sample size N_{eff}	977	960	963
N_{eff}/N	0.49	0.48	0.48
Weighted mass in x_0 basin	58.8%	54.7%	57.7%
Vacuum distribution entropy	0.94 nats	0.97 nats	0.95 nats
Top vacuum probability	58.8%	54.7%	57.7%
Basin exception rate ($d < \ell$)	0.0%	0.0%	0.0%
Weight concentration $\max(w)/\sum w$	1.58×10^{-3}	1.63×10^{-3}	1.58×10^{-3}

- `outputs/bell_locality_summary.csv` — configuration-space locality diagnostics. (Historical filename for configuration-space locality diagnostics.)
- `outputs/wilson_coefficients.csv` — kernel-weighted sector averaging. (Historical filename for kernel-weighted sector averaging.)
- `outputs/ell_sensitivity_results.csv` — ell scan (Exp 9).
- `outputs/ell_sensitivity_summary.json` — aggregated ell metrics.
- `outputs/sigma_ell_boundary_results.csv` — sigma-by-ell grid (Exp 10).
- `outputs/sigma_ell_boundary_summary.json` — aggregated boundary metrics.
- `outputs/landscape_perturbation_results.json` — perturbed-landscape robustness (Exp 12).

3. Paper figures and tables

Figures are generated by `paper/generate_figures.py`. Tables are generated by `paper/generate_tables.py`. The paper is built with `make` in the `paper/` directory.

Appendix D: Degenerate small-ℓ behavior

When the kernel scale ℓ is much smaller than the inter-vacuum spacing ($\ell \lesssim 0.25$), the Gaussian kernel assigns non-negligible weight only to sectors in the x_0 basin. The effective sample size drops to $N_{\text{eff}} \ll N$, and the ISR distribution is effectively a delta function on the x_0 vacuum. KL divergence between successive cutoffs vanishes trivially ($< 10^{-25}$ at $\ell = 0.05$) because there is no distribution to shift.

These values are reported separately here because they are diagnostic artifacts, not evidence of physical stability. Including them alongside non-degenerate results would misleadingly inflate the apparent improvement ratio.

Table XI shows the degenerate rows from the ell sensitivity scan, and Table XII shows the full sigma–ell grid including the small-ℓ columns.

TABLE XI: Degenerate and boundary rows from the ell sensitivity scan (Gaussian kernel). Rows with $\ell \lesssim 0.25$ are degenerate: $N_{\text{eff}} \ll N$, producing trivially vanishing KL drift. The $\ell = 0.23$ row ($N_{\text{eff}}/N = 0.25$) is a boundary case that narrowly fails the $N_{\text{eff}}/N \geq 0.3$ threshold.

ℓ	N_{eff}	KL drift	Wasserstein	Basin exc. rate	Λ_{eff}	Conv. rate
0.05	1.0	$< 10^{-25}$	$< 10^{-125}$	0.00	2.000	—
0.10	49.6	$< 10^{-29}$	$< 10^{-29}$	0.00	2.000	—
0.23	2523	2.0×10^{-10}	1.6×10^{-9}	0.00	2.000	-0.07

TABLE XII: Full sigma–ell boundary grid including degenerate small-ℓ columns. Values at $\ell = 0.10$ and $\ell = 0.25$ are diagnostic artifacts: the narrow kernel forces $N_{\text{eff}} \ll N$, producing trivially large improvement ratios.

$\sigma \setminus \ell$	0.10	0.25	0.50	1.00	2.00
0.5	2.0×10^{21}	5.6×10^{6}	68.5	3.12	1.28
1.0	9.6×10^{20}	2.9×10^{5}	142.5	5.97	1.86
1.5	1.1×10^{21}	4.1×10^{5}	240.0	5.38	2.35
2.0	9.2×10^{20}	2.1×10^{5}	103.1	7.68	4.40
3.0	—	9.5×10^{5}	58.5	6.27	2.46